

HEALTHCARE DATA CLIENT

Data Cleaning & Preparation Methodology

Healthcare Data Quality Automation and Exception Reporting

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Sources	Excel, CSV and SQL/database
Scale	Approximately 250,000 records per batch
Tools	Python, SQL, Excel, Power Query and Power BI

Methodology boundary: The project involved automated data-quality checks across the confirmed validation categories. Exact historical scripts, transformation order, field-level thresholds and source-specific rules are not retained. Detailed preparation steps below are a reproducible portfolio methodology.

1. Preparation Objective

Prepare high-volume healthcare data for consistent validation, exception identification and reporting while preserving record-level traceability. The methodology separates data preparation from exception handling so that invalid or questionable values are not silently changed simply to make a record pass validation.

2. Confirmed Data Quality Checks

Quality Area	Confirmed Check
Uniqueness	Duplicates
Completeness	Nulls and missing values
Validity	Invalid dates, invalid IDs and out-of-range values
Conformance	Incorrect formats
Consistency	Inconsistent codes
Integrity	Cross-field mismatches

3. End-to-End Preparation Flow

Stage	Method
1. Source intake	Receive supported Excel, CSV or SQL/database input and identify the batch context.
2. Structural profiling	Review record count, fields, apparent types, null patterns and formatting issues.
3. Type preparation	Parse and standardize fields only where required for reliable validation.
4. Validation execution	Apply documented data-quality rules to applicable fields.
5. Exception classification	Flag failed records and capture reason codes or failure explanations.
6. Exception output	Create the separate exception file, table or report for targeted review.
7. Reconciliation	Reconcile processed records, exception records and validation failures.
8. Reporting preparation	Prepare governed validation results for Power BI visibility.

4. Source Intake & Profiling

Control	Preparation Method
Batch identification	Maintain a source or batch reference sufficient for reconciliation.
Record population	Establish the processed population before calculating Exception Rate.
Field inventory	Identify available fields and their validation relevance.
Type profiling	Assess dates, identifiers, codes, numeric values and text fields before conversion.
Null profiling	Identify missing-value patterns without fabricating replacement healthcare values.
Format profiling	Identify inconsistent representations that may affect validation.

5. Duplicate Preparation

Duplicate detection is a confirmed validation area. The exact historical duplicate-defining field combination is not retained. For a portfolio rebuild, the duplicate rule must be explicitly documented before execution rather than inferred from healthcare domain assumptions.

Control: Duplicate handling must not inflate Records Processed or Exception Records. The original source relationship should remain traceable.

6. Missing-Value Preparation

Step	Method
Identify	Profile null, blank or missing values in fields subject to completeness validation.
Classify	Determine whether the applicable documented rule treats the missing value as an exception.
Preserve	Do not insert invented patient, claim, provider, clinical or financial values.
Flag	Where the rule fails, retain the record and assign the relevant exception reason.

7. Date & Format Preparation

Area	Method
Dates	Parse source date values using documented rules and flag values that cannot be validly interpreted.
Formats	Compare values with the documented expected structure where a format rule exists.
Identifiers	Preserve identifier semantics and validate against documented ID rules rather than treating identifiers as measures.
Ranges	Evaluate values against documented valid bounds where an applicable range rule exists.

8. Code & Cross-Field Preparation

Area	Method
Code consistency	Standardize or map codes only when a documented valid set or mapping is available.
Cross-field integrity	Evaluate related fields using documented relationships.
Unsupported rules	Do not infer healthcare code sets, clinical relationships or adjudication logic that were not retained from the engagement.
Failure capture	Record the affected source reference and reason for each failed rule.

9. Exception Construction

Element	Purpose
Exception Flag	Indicates whether the record failed one or more applicable rules.
Reason Code	Classifies the validation failure.
Validation Rule	Identifies the check that failed.
Validation Message	Provides reviewer-readable failure context where implemented.
Source Record Reference	Supports reconciliation to the processed record.
Batch Reference	Connects the exception to its file or batch context.

Grain control: One record can fail multiple rules. Record-level exception reporting counts the record once, while a rule-level failure table can contain multiple rows for that record.

10. Reconciliation Controls

Control	Expected Relationship
Processed population	Represents the governed batch population.
Exception population	Subset of processed records that failed at least one applicable rule.
Validation failures	May exceed Exception Records because one record can fail multiple rules.
Exception Rate	Exception Records / Records Processed x 100.
Observed engagement result	2% Exception Rate.
Traceability	Each failure remains attributable to the affected record and validation reason.

11. Tool Responsibilities

Tool	Portfolio Methodology Role
Python	Repeatable validation, profiling and exception-generation logic.
SQL	Data extraction, querying, reconciliation and validation support.
Excel	Source review, spot checks and exception-review support.
Power Query	Reporting-layer preparation and transformation.
Power BI	Data-quality and exception reporting visibility.

Historical boundary: The tools above are confirmed. Their exact historical division of responsibility, scripts and transformation sequence are not retained.

12. Quality Control Matrix

Risk	Control
Duplicate inflation	Apply the documented duplicate rule and reconcile record counts.
Incorrect null treatment	Preserve missing values until the applicable rule determines exception status.
Date conversion errors	Validate parsing before using converted dates.
Invented code mapping	Use only documented code sets or mappings.
Rule-level double counting	Separate record-level exception measures from rule-level failure counts.
Lost traceability	Retain source and batch references through validation output.
Reporting mismatch	Reconcile Power BI measures to governed validation results.
Confidentiality exposure	Exclude PHI and confidential healthcare values from public portfolio artifacts.

13. Preparation Acceptance Criteria

ID	Acceptance Condition
DP-01	The processed population can be reconciled to the governed source batch.
DP-02	Applicable validation fields are interpretable using documented analytical types.
DP-03	Known duplicate, null, date, format, ID, range, code and cross-field checks can be executed.
DP-04	Failed records remain traceable to source and failure reason.
DP-05	Exception Records are not inflated by multiple rule failures on the same record.
DP-06	Exception Rate can be recalculated from governed record-level populations.
DP-07	Reporting-ready outputs do not introduce fabricated healthcare values.

14. Confirmed Outcome Context

Outcome: The engagement processed approximately 250,000 records per batch, surfaced a 2% exception rate and reduced review turnaround from 6 hours to 2.5 hours per file. The methodology supports the validation and exception-review process but does not claim unverified clinical, compliance or cost outcomes.